

Federated Learning Round

1 **Model Broadcast**
The server sends the global model parameters to the clients.

2 **Local Training**
Each client trains the model on its own private local data

3 **Weights Upload**
Each client sends only the trained model weights to the server

4 **Model Update**
The server aggregates trained weights into an updated global model

5 **Agentic Layer**
Reconfiguration of Architectural Pattern

Multiple AI-Agents Layer



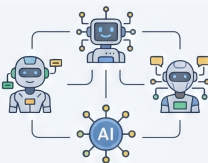
System Evaluation Metrics



System Configuration Parameters



Contextual Prompt



Uses LLMs to decide whether to activate or deactivate **Architectural Patterns** at the end of each FL round

Multiple AI-Agents Reasoning

ON

OFF

ON

Architectural Patterns



Client Selector

Selects clients for the next training round based on predefined criteria



Heterogeneous Data Handler

Mitigates non-IID data effects by generating synthetic training data



Message Compressor

Compresses client-server messages to reduce communication time

Multiple AI-Agents Coordination Mechanisms

AP1 ✓ AP1 ✓ AP1 ✓
AP2 ✗ AP2 ✗ AP2 ✓
AP3 ✓ AP3 ✗ AP3 ✗



Voting-based

Agents independently vote on whether to activate or deactivate each Architectural Pattern



Role-based

Each agent specializes in one architectural pattern and decides whether to (de)activate it



Debate-based

Agents challenge each other's rationales to converge toward a shared decision

Results



Avg. Accuracy +10%

Dynamic Adaptation of Architectural Patterns improves model accuracy



Avg. Round Time -8%

Dynamic Adaptation of Architectural Patterns reduces the average duration of Federated Learning rounds